v1.0

Steel Cutting Optimization & Inventory Management System

📋

(95%+).

- ✅ First Fit Decreasing + Local Search + Dynamic Programming
- ✅
- ✅
- ✅
- ✅ (PDF/Excel)
- ✅
- ✅ SQLite
- ✅ API RESTful

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```
steel-cutting-system/
├── steel_cutting_app.py          # □□□□□□□ □□□□□□□ (Python)
├── advanced_optimizer.py         # □□□□□□□□□□ □□□□□□□□ (DP)
├── api_backend.py                # FastAPI Backend
├── archive_reports.py            # □□□□ □□□□□□□ □□□□□□□□□
├── steel_cutting_ui.html         # □□□□□ □□□□□□□□ □□□□□□□□
├── steel_cutting_ui_advanced.html # □□□□□□□ □□□□□□□□
├── steel_cutting_db.sqlite       # □□□□□ □□□□□□□□
├── technical_documentation.md    # □□□□□□□ □□□□□□□
├── cutting_patterns.png          # □□□ □□□□ □□□□□□□□
├── summary_chart.png             # □□□ □□□□□□ □□□□□□□□
└── README.md                     # □□□ □□□□□
```

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- Python 3.9+
- pip

```
# 1. □□□□□ □□□□ □□□□□□□□ (□□□□□□□□)
python -m venv venv
source venv/bin/activate  # Linux/Mac
# □□
```

```
venv\Scripts\activate  # Windows

# 2. □□□□□ □□□□□□□□□
pip install fastapi uvicorn sqlalchemy matplotlib pandas openpyxl reportlab

# 3. □□□□□ □□□□□□□□
python steel_cutting_app.py

# 4. □□□□□ API (□□□□□□□□)
uvicorn api_backend:app --reload --host 0.0.0.0 --port 8000
```

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	95.4%
	4.6%
	10
	2
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🎯

1.

```
requirements = [
    RequiredPiece("_□□□□A1", 350, 4, "_□□□□12mm"),
    RequiredPiece("_□□□B1", 180, 6, "_□□□□12mm"),
]
```

2.

```
optimizer = CuttingOptimizer(stock_bars, inventory)
result = optimizer.optimize(requirements, use_inventory=True)
```

3.

```
print(f"□□□□ :□□□□□□□□□ {result['stats']['utilization_rate']}%")
```

🔧 API Endpoints

Endpoint	Method
`/api/v1/optimize`	POST
`/api/v1/inventory`	GET

Table 2 – continued

Endpoint	Method
`/api/v1/inventory`	POST
`/api/v1/projects`	GET/POST
`/api/v1/archive/search`	GET
`/api/v1/reports/waste`	GET

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2) 3-4)

- Column Generation
-
- QR Codes

3) 5-6)

- Tekla Structures
- / Excel
-

4) 7+)

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: **:** 2026-05-23 **:** 1.0.0